

Maharashtra HSC Exam, Physics

March 2017

Time: 3 Hours

Total Marks: 70

Note:

- (i) All questions are compulsory.
- (ii) Neat diagrams must be drawn wherever necessary.
- (iii) Figures to the right indicate full marks.
- (iv) Use of only logarithmic tables is allowed.
- (v) All symbols have their usual meaning unless otherwise stated.
- (vi) Answer to both sections must be written in the same answer book.
- (vii) Answer to every question must be written on a new page.

SECTION –I

Question 1. Select and write the most appropriate answer from the given alternatives for each sub-question. [7]

- (i) If the pressure of an ideal gas decreases by 10% isothermally, then its volume will.....
 - (a) Decreases by 9%
 - (b) Increase by 7%
 - (c) Increase by 10%
 - (d) Increase by 11.4%
- (ii) Stretching of a rubber band results in.....
 - (a) No change in potential energy
 - (b) Zero value of potential energy
 - (c) Increase in potential energy
 - (d) Decrease in potential energy
- (iii) When the angular acceleration of a rotating body is zero, which physical quantity will be equal to zero?
 - (a) Angular momentum
 - (b) Moment of inertia
 - (c) Torque
 - (d) Radius of gyration
- (iv) In a damped harmonic oscillator, periodic oscillations have Amplitude.
 - (a) Gradually increase
 - (b) Suddenly increasing

- (c) Suddenly decreasing
- (d) Gradually decreasing

(v) A sine wave of wavelength λ is travelling in a medium. What is the minimum distance between two particles of the medium which always have the same speed?

- (a) λ
- (b) $\frac{\lambda}{2}$
- (c) $\frac{\lambda}{3}$
- (d) $\frac{\lambda}{4}$

(vi) Velocity of transverse wave along a stretched string is proportional to

- (a) $\sqrt{T}$
- (b) T
- (c) $\frac{1}{\sqrt{T}}$
- (d) $\frac{1}{T}$

(vii) Find the wavelength at which a black body radiates maximum energy, if its temperature is 427°C .

- (a) $0.0414 \times 10^{-6} \text{ m}$
- (b) $4.14 \times 10^{-6} \text{ m}$
- (c) $41.4 \times 10^{-6} \text{ m}$
- (d) $414 \times 10^{-6} \text{ m}$

Solution 1:

- (i) (d) Increase by 11.4%
- (ii) (c) Increase in potential energy
- (iii) (c) Torque
- (iv) (d) Gradually decreasing
- (v) (b) $\frac{\lambda}{2}$
- (vi) (a) $\sqrt{T}$
- (vii) (b) $4.14 \times 10^{-6} \text{ m}$

Question 2: Attempt any six.

(i) Explain the concept of centripetal force.

(ii) Prove that root mean square velocity of gas molecule is directly proportional to the square root of its absolute temperature.

(iii) Obtain the differential equation of linear simple harmonic motion.

(iv) Draw a neat, labeled diagram for a liquid surface in contact with solid, when the angle of contact is acute.

(v) A hole is drilled half way to the center, of the Earth. A body is dropped into the hole. How much will it weight at the bottom of the hole if the weight of the body on the Earth's surface is 350 N?

(vi) A solid sphere of mass 1 kg rolls on a table with linear speed 2m/s, find its total kinetic energy.

(vii) A transverse wave, when is produced on a stretched string 0.9 m long and fixed at its ends. Find the speed of the transverse wave, when the string vibrates while emitting second overtone of frequency 324 Hz.

(viii) A body cools at the rate of $0.5^{\circ}\text{C}/\text{minute}$ when it is 25°C above the surroundings. Calculate the rate of cooling when it is 15°C above the same surroundings.

Solution 2:

(i) The force which is required to move a body uniformly in a circle. It acts along the radius and towards the center of the circle. It is also known as centripetal force. IF m is the mass of the particle, so the magnitude of the centripetal force is.

$$\frac{mv^2}{r} = m\omega^2 r$$

(ii)

$$P = \frac{1}{3} \frac{mn}{V} \times \bar{v}^2$$

$$\Rightarrow PV = \frac{1}{3} mn \times \bar{v}^2$$

$$\because PV = RT \text{ and } mn = M$$

$$\Rightarrow \bar{v}^2 = \frac{3RT}{M}$$

$$\Rightarrow V_{rms} = \sqrt{\bar{v}^2}$$

$$\Rightarrow V_{rms} = \sqrt{\frac{3RT}{M}}$$

$$\Rightarrow sV_{rms} \propto \sqrt{T}$$

(iii) Let us take a particle performs linear SHM. The force on the particle is always directed towards the mean position. If the force acting on the particle is F at displacement $\vec{x}$ from the mean position.

$$\vec{F} = -K\vec{x} \dots\dots\dots 1$$

Where, K is the force constant, and –ve sign indicates the force and displacement are in opposite direction.

So,

$$\text{Velocity of a particle} = \frac{d\vec{x}}{dt}$$

Acceleration

$$= \frac{d^2\vec{x}}{dt^2}$$

Therefore,

Force is equal to the product of mass and acceleration,

$$\vec{F} = m \times \frac{d^2\vec{x}}{dt^2} \dots\dots\dots 2$$

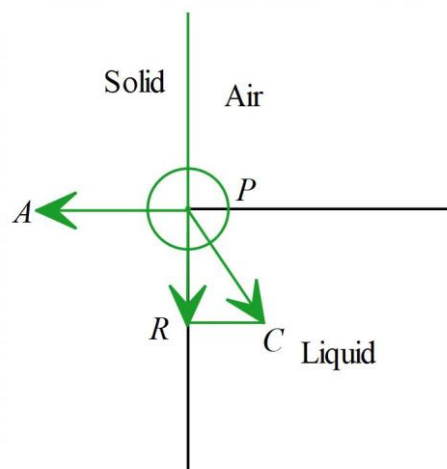
From eq 1 and 2 we get

$$m \times \frac{d^2\vec{x}}{dt^2} = -K\vec{x}$$

$$\Rightarrow \frac{d^2\vec{x}}{dt^2} + \frac{K}{m} \vec{x} = 0$$

Hence, the required expression is $\frac{d^2\vec{x}}{dt^2} + \frac{K}{m}\vec{x} = 0$.

(iv)



Here, a molecule P in liquid surface is in contact with solid. The forces on molecules P-

- (a) Solid-liquid adhesive force $\vec{PA}$
- (b) Liquid-air cohesive force $\vec{PC}$
- (c) Negligible liquid-air adhesive force.
- (d) Negligible weight of molecule acting vertically downward.
- (v) We have,

$$d = 3200\text{km}, R = 6400\text{km}, g = 9.8\text{m/s}^2$$

$$W = 350\text{N}$$

Weight at bottom of hole

$$\begin{aligned} W_d &= W \left(\frac{R-d}{R} \right) \\ &= 350 \left(\frac{6400-3200}{6400} \right) \\ &= 350 \times \frac{3200}{6400} \\ &= 175\text{N} \end{aligned}$$

(vi)

$$M = 1\text{kg}, V = 2\text{m/s}.$$

We know that the total kinetic energy is equal to the sum of translational energy and rotational energy

$$\text{Total K.E} = \frac{1}{2}mv^2$$

$$= \frac{1}{2} \times 1 \times 2 \times 2$$

$$= 2 \text{ Joule}$$

Hence, total kinetic energy is 2 Joule .

(viii) The length of wire = 0.9m

Frequency, $\eta = 324 \text{ Hz}$

Speed of the transverse waves

$$v = n\lambda$$

$$= n(2l)$$

$$= 324 \times 2 \times 0.9$$

$$= 583.2 \text{ m/s}$$

Hence, the speed of the transverse wave is 583.2 m/s .

(viii) We have,

$$\left(\frac{d\theta}{dt} \right) = 0.5^\circ \text{ C / min}$$

$$\theta_1 - \theta_2 = 25^\circ \text{ C}, \theta_2 - \theta_1 = 15^\circ \text{ C}$$

So, by the Newton law of cooling

$$\begin{aligned}\left(\frac{d\theta}{dt}\right) &= k(\theta - \theta_0) \\ \Rightarrow \frac{\left(\frac{d\theta}{dt}\right)_1}{\left(\frac{d\theta}{dt}\right)_2} &= \frac{(\theta_1 - \theta_0)}{(\theta_2 - \theta_0)} \\ \Rightarrow \frac{\left(\frac{d\theta}{dt}\right)_1}{\left(\frac{d\theta}{dt}\right)_2} &= \frac{25^0}{15^0} \\ \Rightarrow \left(\frac{d\theta}{dt}\right)_2 &= \frac{25^0}{15^0} \times \left(\frac{d\theta}{dt}\right)_1 \\ \Rightarrow \left(\frac{d\theta}{dt}\right)_2 &= \frac{25^0}{15^0} \times 0.5 \\ &= \frac{7.5}{25} \\ &= \left(\frac{d\theta}{dt}\right)_2 = 0.3^0 C / \text{min}\end{aligned}$$

Hence, the rate of cooling when it is $15^0 C$ above the same surroundings is $0.3^0 C / \text{min}$

Question 3: Attempt any THREE:

[9]

- (i) Show that the period of a satellite revolving around the Earth depends upon mass of the Earth.**
- (ii) Obtain an expression for torque acting on a rotating body with constant angular acceleration. Hence state the dimensions and SI unit of torque.**
- (iii) The total energy of free surface of a liquid drop is 2π times the surface tension of the liquid. What is the diameter of the drop? [Assume all terms in SI unit]**
- (iv) A vehicle is moving on a circular track whose surface is inclined towards the horizon at an angle of 10^0 . The maximum velocity with which it can move safely is 36 km/hr . calculate the length of the circular track.**

Solution 3:

(i) We know that the critical speed of the circular orbit is $V_c = \sqrt{\frac{GM}{r}}$.

Where, G is the gravitational constant, M is the mass of the earth, r is the radius of the circular orbit and R is the radius of the Earth, h is the altitude of an orbit.

Therefore, in time T the satellite travels:

Distance is equal to the circumference of its circular orbit

$$T = \frac{\text{Circumference}}{\text{Critical speed}}$$

$$= \frac{2\pi r}{V_c}$$

$$= \frac{2\pi r}{\sqrt{\frac{GM}{r}}}$$

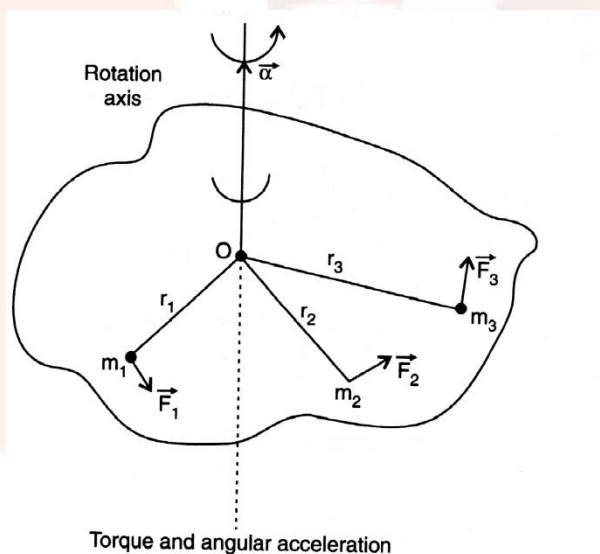
$$= 2\pi \sqrt{\frac{r^3}{GM}}$$

$$= 2\pi \sqrt{\frac{(R+h)^2}{GM}}$$

Hence, the required expression for period of revolution of satellite which clearly shows T is

depend on the mass of planet is $2\pi \sqrt{\frac{(R+h)^2}{GM}}$.

(ii)



Torque

$$\begin{aligned}
 \vec{\tau}_1 &= \vec{r}_1 \times \vec{F}_1 \\
 F_1 &= m_1 a_1 = m_1 r_1 \alpha \\
 \vec{\tau}_1 &= r_1 (m_1 r_1 \alpha) \\
 &= m_1 r_1^2 \alpha \\
 \vec{\tau}_2 &= m_2 r_2^2 \alpha, \\
 \tau_3 &= m_3 r_3^2 \alpha, \\
 \tau_n &= m_N r_N^2 \alpha, \\
 \tau &= \vec{\tau}_1 + \vec{\tau}_2 + \dots + \vec{\tau}_N \\
 &= m_1 r_1^2 \alpha + m_2 r_2^2 \alpha + m_3 r_3^2 \alpha + \dots + m_N r_N^2 \alpha \\
 &= \left(\sum_{i=1}^N m_i r_i^2 \right) \alpha
 \end{aligned}$$

Where, I is the moment of inertia.

(iii) We have,

$$4\pi r T^2 = 2\pi T$$

$$2r^2 = 1$$

$$r^2 = \frac{1}{2}$$

$$r = \frac{1}{\sqrt{2}}$$

$$= \frac{1}{1.414}$$

$$d = 2r$$

$$= 2 \times \frac{1}{1.414}$$

$$= 1.4144m$$

Hence, the diameter of the drop is 1.4144m.

(iv)

We have,

$$\theta = 10^\circ, \tan 10^\circ = 0.6484$$

$$g = 9.8m/s^2, r = ?, v = 36km/hr = 10m/s$$

Therefore,

$$v = \sqrt{rg \tan \theta}$$

$$10 = \sqrt{r \times 9.8 \times 0.6484}$$

$$(10)^2 = r \times 9.8 \times 0.6484$$

$$r = \frac{10 \times 10}{9.8 \times 0.6484}$$

$$= 15.74m$$

Hence, the length of the circular track is 15.74m.

Question 4: Prove the law of conservation of energy for a particle performing simple harmonic motion. Hence graphically show the variation of kinetic energy and potential energy w.r.t instantaneous displacement.

Two sound notes have wavelengths $\frac{83}{170}$ m and $\frac{83}{172}$ m in the air. These notes when sounded together produce 8 beats per second. Calculate the velocity of sound in the air and frequencies of the two notes.

OR

Explain the formation of stationary waves by analytical method. Show the formation of stationary wave diagrammatically.

A mass of 1 kg is hung from a steel wire of radius 0.5 mm and length 4m. Calculate the extension produced. What should be the area of cross section of the wire so that elastic limit is not exceeded? Change in radius is negligible (Given: $g = 9.8m/s^2$ Elastic limit of steel $2.4 \times 10^8 N/m^2$ for steel (Y_{steel}) = $20 \times 10^{10} N/m^2$, $\pi = 3.142$).

Solution 4:

Let us consider a particle having mass m and amplitude A, at a distance x from, the mean position:

$$\text{Potential energy: } PE = \frac{1}{2} kx^2 = \frac{1}{2} m\omega^2 x^2$$

$$\begin{aligned} \text{Kinetic energy: } KE &= \frac{1}{2} k(A^2 - x^2) \\ &= \frac{1}{2} m\omega^2 (A^2 - x^2) \end{aligned}$$

$$\text{Where K is the constant, } \omega = \sqrt{\frac{K}{m}}$$

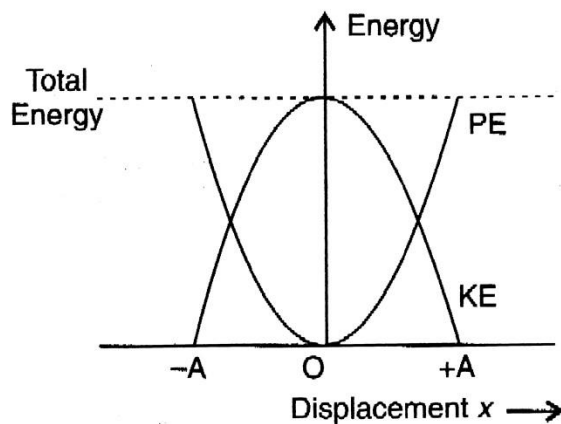
Total energy is equal to the sum of potential and kinetic energy

$$E = PE + KE$$

$$= \frac{1}{2} m \omega^2 x^2 + \frac{1}{2} m \omega^2 (A^2 - x^2)$$

$$= \frac{1}{2} m \omega^2 A^2$$

Which shows the total energy is independent of distance x and time t .



Variation of KE & PE with respect to instantaneous x .

Given:

$$\lambda_1 = \frac{83}{170} m, \lambda_2 = \frac{83}{172} m.$$

Let us suppose the corresponding frequencies be n_1 and n_2

$$v = n_1 \lambda_1 = n_2 \lambda_2$$

Therefore, velocity $n_1 = \frac{v}{\lambda_1}, n_2 = \frac{v}{\lambda_2} \dots\dots 1$

Now,

$$\lambda_1 > \lambda_2$$

$$\Rightarrow n_2 > n_1$$

So, from eq(1) we get,

$$n_2 - n_1 = 8$$

$$\Rightarrow v \left(\frac{1}{\lambda_2} - \frac{1}{\lambda_1} \right) = 8$$

$$\Rightarrow v \left(\frac{172}{83} - \frac{170}{83} \right) = 8$$

$$\Rightarrow v \left(\frac{2}{83} \right) = 8$$

$$\Rightarrow v = \frac{8 \times 83}{2}$$

$$= 332 \text{ m/s}$$

Therefore,

$$n_1 = \frac{v}{\lambda_1}$$

$$= 332 \times \frac{173}{83}$$

$$= 680 \text{ Hz}$$

$$n_2 = \frac{v}{\lambda_2}$$

$$= 332 \times \frac{172}{83}$$

$$= 688 \text{ Hz}$$

OR

Let us consider the two simple harmonic progressive waves of equal amplitude and frequency propagating on a long uniform string in opposite direction i.

If the wave frequency is 'n' and wavelength 'l' is travelling along X-axis, then

$$Y_1 = a \sin 2\pi \left(nt - \frac{x}{\lambda} \right)$$

If the wave frequency is 'n' and wavelength 'l' is travelling along negative X-axis, then

$$Y_2 = a \sin 2\pi \left(nt + \frac{x}{\lambda} \right)$$

These waves interfere to produce stationary waves.

$$Y = Y_1 + Y_2$$

$$= a \sin 2\pi \left(nt - \frac{x}{\lambda} \right) + a \sin 2\pi \left(nt + \frac{x}{\lambda} \right)$$

By using

$$\sin C + \sin D = 2 \sin \left(\frac{C+D}{2} \right) \cos \left(\frac{C-D}{2} \right)$$

$$\text{and } \cos(-\theta) = \cos \theta$$

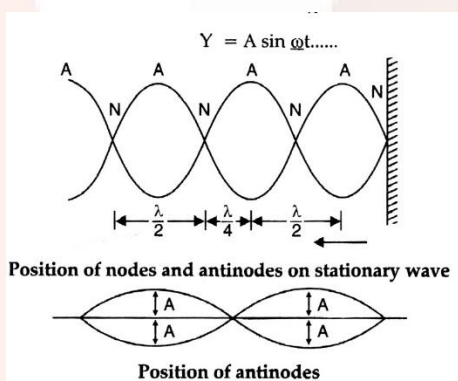
$$Y = a \sin 2\pi \left(nt - \frac{x}{\lambda} \right) + a \sin 2\pi \left(nt + \frac{x}{\lambda} \right)$$

$$= 2a \left[\sin(2\pi nt) \cos \left(\frac{2\pi x}{\lambda} \right) \right]$$

$$= \left(2a \cos \frac{2\pi x}{\lambda} \right) \sin(2\pi nt)$$

$$Y = A \sin(2\pi nt)$$

Where, A is the amplitude of resultant wave $A = \left(2a \cos \frac{2\pi x}{\lambda} \right)$



Here, the antinodes is maximum,

$$A = \left(2a \cos \frac{2\pi x}{\lambda} \right)$$

$$\pm 2 = 2 \cos \frac{2\pi x}{\lambda}$$

$$\cos \frac{2\pi x}{\lambda} = \pm 1$$

$$\frac{2\pi x}{\lambda} = 0, \pi, 2\pi, \dots$$

$$\frac{2\pi x}{\lambda} = p\pi$$

$$x = \frac{p\lambda}{2}$$

$$\text{So, for } x=0, \frac{\lambda}{2}, \lambda, \frac{3\lambda}{2}, \dots$$

Therefore, distance between two successive antinodes is $\frac{\lambda}{2}$.

Amplitude of nodes is zero, $A=0$

$$A = 2a \cos \frac{2\pi x}{\lambda}$$

$$0 = 2a \cos \frac{2\pi x}{\lambda}$$

$$\cos \frac{2\pi x}{\lambda} = 0$$

$$\frac{2\pi x}{\lambda} = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \dots$$

$$\text{So, for } x=0, \frac{\lambda}{4}, \frac{3\lambda}{4}, \dots$$

Given:

Radius of wire $0.5\text{mm} = 5 \times 10^{-4}\text{m}$

Length is 4m, $M = 1\text{kg}$ $g = 9.8\text{m/s}^2$, $Y = 20 \times 10^{10}\text{N/m}^2$, $\pi = 3.142$

Therefore,

$$\begin{aligned}
 Y &= \frac{MgL}{\pi r^2 \Delta L} \\
 \Delta L &= \frac{MgL}{\pi r^2 \Delta Y} \\
 &= \frac{1 \times 9.8 \times 4}{3.142 \times (5 \times 10^{-4})^2 \times 20 \times 10^{10}} \\
 &= \frac{9.8 \times 4 \times 10^{-2}}{3.142 \times 25 \times 20} \\
 &= 0.2495 \times 10^{-3} \text{ m} \\
 &= 0.2495 \text{ mm}
 \end{aligned}$$

Now, Area of the cross-section of the wire

$$\begin{aligned}
 \pi r^2 &= 3.142 \times (5 \times 10^{-4})^2 \\
 &= 3.142 \times 25 \times 10^{-8} \\
 &= 7.855 \times 10^{-7} \text{ m}^2
 \end{aligned}$$

SECTION -II

Question 5: Select and write the most appropriate answer from the given alternative for each sub-questions: [7]

(i) If A.C voltage is applied to pure capacitor, then voltage across the capacitor.....

- (a) Leads the current by phase angle $\left(\frac{\pi}{2}\right)$ rad.**
- (b) Leads the current by phase angle (π) rad.**
- (c) Lags behind the current by phase angle $\left(\frac{\pi}{2}\right)$ rad.**
- (d) Lags behind the current by phase angle (π) rad.**

(ii) In Doppler effect of light, the terms “red shift” is used for.....

- (a) Frequency increase**
- (b) Frequency decrease**
- (c) Wavelength decrease**
- (d) Frequency and wavelength increase.**

(iii) If a watch-glass containing a small quantity of water is placed on two dissimilar magnetic poles, then water

- (a) Shows a depression in the middle.
- (b) Shows an elevation in the middle.
- (c) Surface remains horizontal
- (d) Frequency and wavelength increase.

(iv) Any device that converts one form of energy into another is termed as

- (a) Amplifier
- (b) Transducer
- (c) Receiver
- (d) Demodulator

(v) When a p-n-p transistor is operated in saturation region, then it's.....

- (a) Base-emitter junction is forward biased and base-collector junction is reverse biased.
- (b) Both base-emitter and base-collector junctions are forward biased.
- (c) Both base-emitter and base-collector junction are forward biased.
- (d) Base-emitter junction is reverse biased and base-collector junction is forward biased.

(vi) In a photon- electron collision.....

- (a) Only total energy is conserved.
- (b) Only total momentum is conserved.
- (c) Both total energy and total momentum are conserved.
- (d) Both total momentum and total energy are not conserved.

(vii) If the charge on the condenser of ($10\mu F$) is doubled. Then the energy stored in it becomes.....

- (a) Zero
- (b) Twice that of initial energy
- (c) Half the initial energy.
- (d) Four times the initial energy

Solution 5:

(i) (c) Lags behind the current by phase angle $\left(\frac{\pi}{2}\right)$ rad

(ii) (b) Frequency decrease

(iii) (a) Shows a depression in the middle

(iv) (b) Transducer

(v) (c) Both base-emitter and base-collector junction are forward biased

(vi) (c) Both total energy and total momentum are conserved.

(vii) (b) Twice that of initial energy

Question 6: Attempt any SIX:

[12]

- (i) Distinguish between the phenomenon of interference and diffraction of light.
- (ii) Explain how moving coil galvanometer is converted in to voltmeter. Derive the necessary formula.
- (iii) State the advantage of potentiometer over voltmeter.
- (iv) Draw a neat, labeled block diagram of receiver for the detection of amplitude modulated wave.
- (v) A rectangular coil of a moving coil galvanometer contains 100 turns 15cm^2 . It is suspended in the radial magnetic field 0.03T . The twist constant suspension fiber is $15 \times 10^{-10} \text{ Nm / deg ree}$. Calculate the sensitivity of the moving coil galvanometer.
- (vi) The magnetic flux through a loop is varying according to a relation $\phi = 6t^2 + 7t + 1$. Where ϕ is in milliweber and t is in second. What is the e.m.f induced in the loop at $t = 2\text{s}$.
- (vii) An unknown resistance is placed in the left gap and resistance of 50 ohm is placed in the right gap of a meter bridge. The null point is obtained as 40 cm from the left end. Determine the unknown resistance.
- (viii) Find the frequency of revolution of an electron in Bohr's 2nd orbit ; if the radius and speed of electron in that orbit is $2.14 \times 10^{-10} \text{ m}$ and $1.09 \times 10^6 \text{ m / s}$ respectively. $[\pi = 3.142]$.

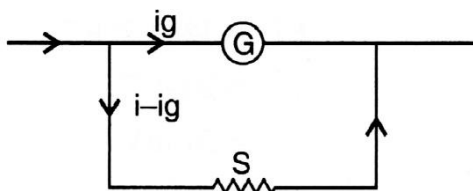
Solution 6:

(i) The difference between the phenomenon of interference and diffraction of light are as follows:

S.No	Interference	Diffraction
1	The superposition of a few coherent waves.	The superposition at a point involves a large number of waves coming from different parts of the same wave front.
2	Double-slit interference fringes are all equal width.	The non-central maxima are of equal width which is half of that of central maximum in single-slit diffraction.
3	Bright fringes are of equal intensity in double slit interference.	Successive non-central maxima decrease rapidly in the intensity in diffraction.
4	The bright and dark fringes are equally spaced in double-slit interference.	Only the central maxima lie approximately halfway between minima in diffraction.

(ii) If a low resistance is connected in parallel in moving coil galvanometer is converted in voltmeter.

Let us consider 'G' is the resistance of galvanometer coil and I_g is maximum current and also connect 'G' and 'S' in parallel.



$$I_g \times G = (i - i_g) S$$

$$\Rightarrow I_g \times G = i_s - i_g S$$

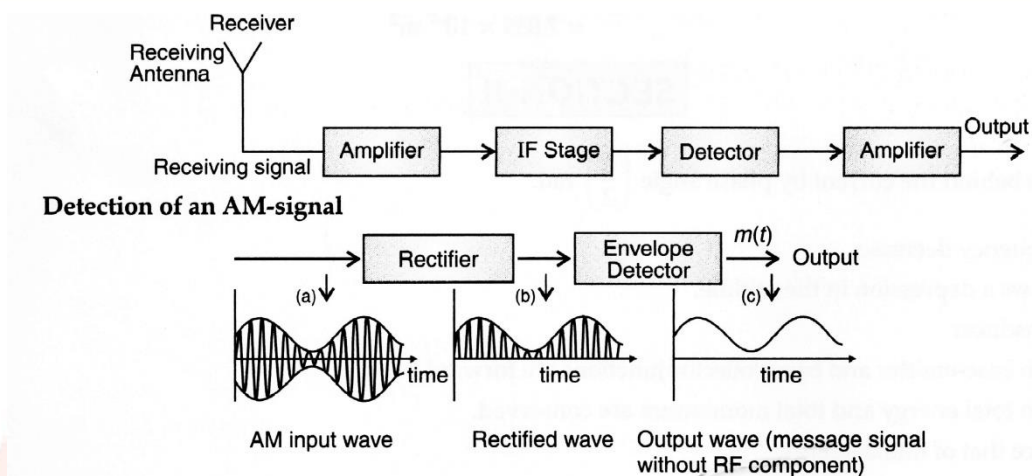
$$\frac{i_g}{i} = \frac{S}{S + G}$$

$$S = \frac{i_g G}{i - i_g}$$

(iii) The advantages of potentiometer over voltmeter are as follows:

Voltmeter	Potentiometer
It is the instrument which measure electric potential.	It is an instrument which measure electromotive force by balancing it against the potential difference.
It has very high value of resistance	It is based on no deflection.
Voltmeter can only measure terminal voltage of given coil.	When it gives zero deflection then it does not draw any current.

(iv)



(v) We have,

$$N = 100, A = 15\text{cm}^2 = 15 \times 10^{-4}\text{m}^2$$

$$B = 0.03\text{T}, C = 15 \times 10^{-10}\text{Nm / deg ree}$$

$$\begin{aligned} S_i &= \frac{NAB}{C} \\ &= \frac{100 \times 15 \times 10^{-4} \times 0.03}{15 \times 10^{-10}} \\ &= 3 \times 10^6 \text{ div / A} \end{aligned}$$

(vi) We have,

$$\text{Magnetic flux, } \phi = 6t^2 + 7t + 1\text{mWb}, t = 2\text{s}$$

Therefore, Induced emf ,

$$\begin{aligned} |E| &= \frac{d\phi}{dt} \\ &= \frac{d(6t^2 + 7t + 1)}{dt} \\ &= 12t + 7 \\ \therefore t &= 2\text{s} \\ |E| &= 12 \times 2 + 7 \\ &= 24 + 7 \\ &= 31\text{mV} \end{aligned}$$

(vii) We have,

$$R = 50\Omega, L_X = 40cm$$

$$L_R = 100 - L_X = 60cm$$

Therefore, the resistance X in the left gap and R in the right gap is $R = 50\Omega$.

$$\frac{X}{R} = \frac{L_X}{L_R}$$

$$\Rightarrow X = R \frac{L_X}{L_R}$$

$$\Rightarrow X = 50 \times \frac{40}{60}$$

$$\Rightarrow X = 33.33\Omega$$

(viii) We have,

$$r = 2.14 \times 10^{-10} m$$

$$v = 1.09 \times 10^6 m/s$$

$$v = \omega r$$

$$= 2\pi f r$$

$$f = \frac{v}{2\pi r}$$

$$= \frac{1.09 \times 10^6}{2 \times 3.142 \times 2.14 \times 10^{-10}}$$

$$= 8.105 \times 10^{14} Hz$$

Question 7: Attempt any THREE:

[9]

(i) Explain with neat diagram, how a p-n junction diode is used as a half wave rectifier.

(ii) Explain self-induction and mutual induction.

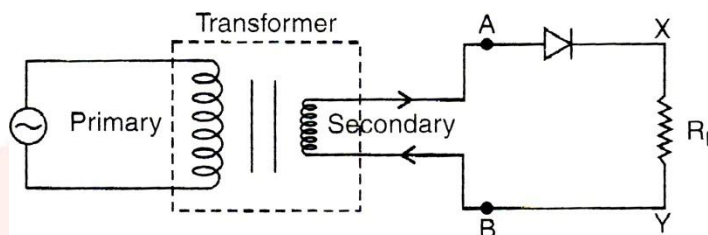
(iii) A cube of marble having each side 1cm is kept in electric field of intensity $300V/m$. Determine the energy contained in the cube of dielectric constant 8.

$$\left[\begin{array}{l} \text{Given} \\ \epsilon_0 = 8.85 \times 10^{-12} C^2 / Nm^2 \end{array} \right]$$

(iv) An electron in an atom revolves around the nucleus in an orbit of radius $0.5\text{\AA}$. IF the frequency of the revolution of an electron is $9 \times 10^9 MHz$, calculate the orbital angular momentum. [charge on an electron is $1.6 \times 10^{-19} C$ and gyromagnetic ratio $8.8 \times 10^{10} C / Kg, \pi = 3.142$]

Solution 7:

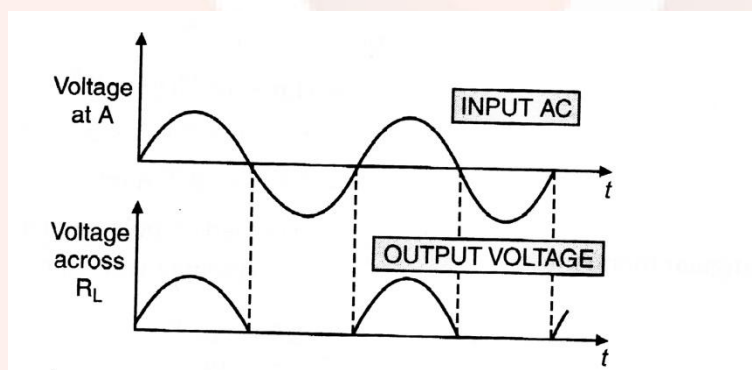
(i)



Working:

During the half input AC, the diode is forward biased and the current flows through R_L . While, during the other half of the input AC, the diode is reverse biased and no current flows through the load R_L .

Such that the given input is rectified.



(ii) **Self-induction:-** Self-induction is defined as the induced emf in the circuit, when the current in the same circuit. For example,

$$\phi \propto I$$

$$\phi = LI$$

Where, L is the proportionality constant and the unit of L is Henry(H).

Mutual induction:- Mutual induction is defined as the induced emf in the circuit when the current in the neighboring circuit changes. For example,

Let us consider the spontaneous current in the primary coil is I . Also, suppose the magnetic flux linked with the secondary coil be ϕ . i.e

$$\phi \propto I$$

$$\phi = MI$$

Where, M is the proportionality constant. It is also called the coefficient of mutual induction .

$$\text{Unit of M is } \frac{V}{AS^{-1}} = VA^{-1}S^{-1}.$$

(iii) We have,

$$l = 1cm = 0.01m$$

$$E = 300V / m, K = 8$$

$$\epsilon_0 = 8.85 \times 10^{-12} C^2 / Nm^2$$

Therefore, energy density in an electrostatic field:

$$= \frac{1}{2} \epsilon_0 KE^2$$

$$= \frac{1}{2} \times 8.85 \times 10^{-12} \times 8 \times (300)^2$$

$$= 8.85 \times 10^{-8} \times 4 \times 9$$

$$= 3.186 \times 10^{-6} J / m^3$$

Volume of cube

$$= l^3 = (0.01)^3 = 1 \times 10^{-6} m^3$$

$\therefore$ energy contained in cube

Energy density $\times$ volume

$$3.186 \times 10^{-6} \times 1 \times 10^{-6}$$

$$= 3.186 \times 10^{-12}$$

(iv) We have,

$$r = 0.5 \text{ \AA} = 5.3 \times 10^{-11} m$$

$$f = 9 \times 10^9 \text{ MHz}, -e = 1.6 \times 10^{-19} C,$$

Gyromagnetic ratio,

$$8.8 \times 10^{10} C / Kg, \pi = 3.142$$

$$\text{Equivalent current, } I = \frac{e}{T} = ef$$

Equivalent magnetic dipole moment:

$$\begin{aligned}
 M &= IA = ef (\pi r^2) \\
 &= (1.6 \times 10^{-19})(9 \times 10^9)(3.142)(5.3 \times 10^{-12})^2 \\
 &= 1.6 \times 9 \times 3.142 \times 5.3 \times 10^{-32} \\
 &= 2.398 \times 10^{-32} \text{ A.m}^2
 \end{aligned}$$

Now,

Orbital angular momentum,

$$\begin{aligned}
 L &= \frac{\text{Magnetic dipole moment}}{\text{Gyromagnetic ratio}} \\
 &= \frac{2.398 \times 10^{-32}}{8.8 \times 10^{10}} \\
 &= 2.725 \times 10^{-41}
 \end{aligned}$$

Question 8:

[7]

Describe the biprism experiment to find the wavelength of the monochromatic light. Draw the necessary ray diagram.

The b width of plane incident wave front is found to be doubled on refraction in denser medium. If it makes an angle of 65° with the normal, calculate the refractive index for the denser medium.

Or

Draw a neat labeled energy level diagram for H atom showing the transition. Explain the series of spectral lines for H atom, whose fixed inner orbit numbers are 3 and 4 respectively.

The work function for potassium and caesium are 2.25 eV and 2.14eV respectively. Is the photoelectric effect possible for either of them if the incident wavelength is $5180\text{\AA}$?

Solution 8:

Distance between the source and eyepiece is directly measured on the scale marked on optical bench. And the micrometer screw fitted to eyepiece is adjusted. And to measure the fringe width β reading of corresponding micrometer is noted. The vertical wire is coincide with 5^{th} , 10^{th} and 15^{th} bright fringes.

Distance d between coherent vertical sources-Conjugate foci method:

Convex length having the focal length less than $d/4$ is mounted on the stand between eyepiece and bi prism.

For position L_1 of lens:

The distance d_1 between two magnified images is measured by micrometer.

For lens L_2 :

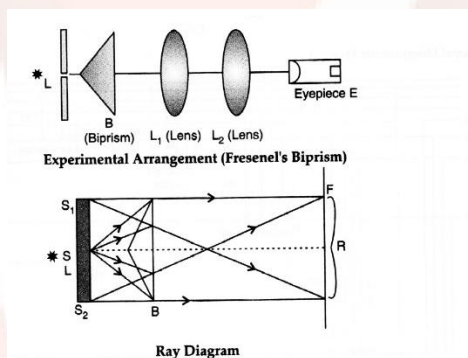
The distance d_2 diminished is measured by moving the lens towards the eyepiece.

Then,

$$d = \sqrt{d_1 d_2}$$

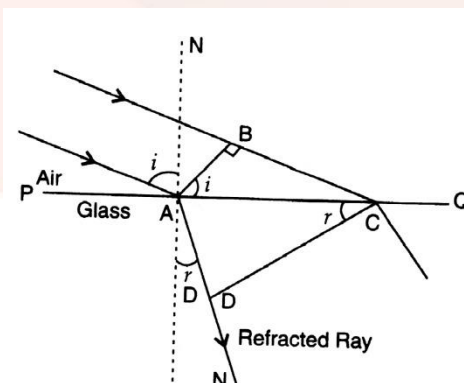
Therefore, the wavelength is

$$\begin{aligned}\lambda &= \frac{\beta d}{D} \\ &= \frac{\beta \sqrt{d_1 d_2}}{D}\end{aligned}$$



We have,

$$i = 65^\circ, \frac{CD}{AB} = 2$$



$$\frac{\cos r}{\cos i} = \frac{CD}{AB} = 2$$

$$\cos r = 2 \cos 65^\circ$$

$$= 2 \times 0.4226$$

$$r = \cos^{-1}(0.8252)$$

$$= 34.91^\circ$$

$$\text{Refractive index} = \frac{\sin i}{\sin r}$$

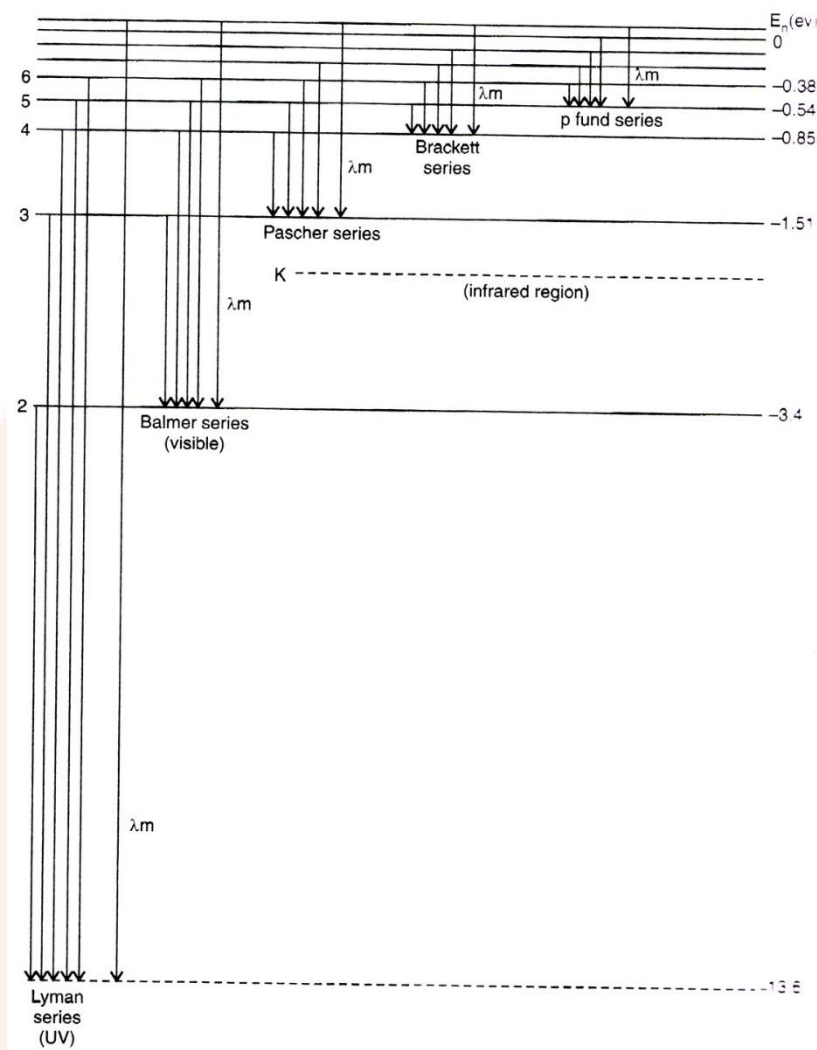
$$= \frac{\sin 65^\circ}{\sin 34.91^\circ}$$

$$= \frac{0.9063}{0.5648}$$

$$= 1.605$$

$$\left[\text{Given: Planck's constant} = 6.63 \times 10^{-34} \text{ Js, Velocity of light} = 3 \times 10^8 \text{ m/s, } V = 1.6 \times 10^{-19} \right]$$

OR



(i) For Lyman series:

Transition

$$-n_f = 1 \text{ and } n_i = 2, 3, 4, \dots$$

$$\text{Wavelength, } \frac{1}{\lambda} = R_H \left[\frac{1}{1^2} - \frac{1}{n_i^2} \right]$$

(ii) For Balmer series:

Transition

$$-n_f = 2 \text{ and } n_i = 3, 4, 5, \dots$$

$$\text{Wavelength, } \frac{1}{\lambda} = R_H \left[\frac{1}{2^2} - \frac{1}{n_i^2} \right]$$

(iii) For Paschan series:

Transition

$$-n_f = 3 \text{ and } n_i = 3, 4, 5, 6, \dots$$

Wavelength,

$$\frac{1}{\lambda} = R_H \left[\frac{1}{3^2} - \frac{1}{n_i^2} \right]$$

(iv) For Brackett series:

Transition

$$-n_f = 4 \text{ and } n_i = 4, 5, 6, 7, \dots$$

Wavelength,

$$\frac{1}{\lambda} = R_H \left[\frac{1}{4^2} - \frac{1}{n_i^2} \right]$$

Transition

$$-n_f = 5 \text{ and } n_i = 5, 6, 7, 8, 9, \dots$$

Wavelength,

$$\frac{1}{\lambda} = R_H \left[\frac{1}{5^2} - \frac{1}{n_i^2} \right]$$

Such that, it lies in the infrared region.

$$\phi(\text{potassium}) = 2.25 \text{ eV}, \phi(\text{caesium}) = 2.14 \text{ eV}$$

$$\lambda = 5180 \text{ \AA} = 5.180 \times 10^{-7} \text{ m}, h = 6.63 \times 10^{-34} \text{ JS}$$

$$\phi(\text{potassium}) = 2.25 \times 1.6 \times 10^{-19}$$

$$= 3.6 \times 10^{-19} \text{ Joule}$$

$$\phi(\text{caesium}) = 2.14 \times 1.6 \times 10^{-19}$$

$$= 3.424 \times 10^{-19} \text{ Joule}$$

$$\text{Now, } v = \frac{c}{\lambda} = \frac{3 \times 10^8}{5.180 \times 10^{-7}} = 0.579 \times 10^{15} \text{ Hz}$$

$$v_0 = \frac{\phi_0}{h} = \frac{3.6 \times 10^{-19}}{6.63 \times 10^{-34}} = 0.543 \times 10^{15} \text{ Hz}$$

$$\nu_0' = \frac{\phi_0'}{h} = \frac{3.424 \times 10^{-19}}{6.63 \times 10^{-34}} = 0.516 \times 10^{15} \text{ Hz}$$

As, $\nu > \nu_0$ and $\nu > \nu_0'$

Hence, photoelectric effect is possible for both potassium as well as caesium.

